



Projecr (signal)

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1. Introduction

Sleep stage classification is an important task in biomedical engineering and healthcare because it helps in monitoring sleep quality and diagnosing sleep disorders. Different physiological signals such as EEG, EMG, and EOG can be used to identify sleep stages.

In this project, EOG-R (Right Electrooculography) signals were analyzed using deep learning techniques to classify different sleep stages automatically. The dataset was preprocessed, segmented into epochs, and balanced using data augmentation methods. A Convolutional Neural Network (CNN) model was then developed and trained to learn patterns from the EOG-R signals and classify them into five sleep stages.

The main goal of this project is to improve the accuracy of automatic sleep stage classification using signal processing and deep learning methods.

2. Installing required library

In this step, the MNE library was installed using pip. MNE is used for processing and analyzing biomedical signals such as EEG and EOG.

3. Data loading and preprocessing

EDF signal files were loaded, and only the EOG-R channel was selected. A filter was then applied to remove noise from the signals. After that, the signals were divided into 30-second segments called epochs.

The XML files were also read to extract sleep stages and convert them into numerical labels. Finally, the processed data and labels were saved for training.

4. Dataset splitting

The dataset was divided into:

- 70% Training
- 15% Validation
- 15% Testing

This was done to ensure proper model training and evaluation.

5. Class Weighting

- Applied class weights to handle class imbalance
- Higher weights assigned to minority classes
- Helps improve classification of underrepresented sleep stages

6. Optimizer selection

The Adam optimizer was used with a learning rate of 0.0003 to improve the model's performance during training.

7. CNN Model Architecture

A CNN model was built using Conv1D layers to extract features from EOG-R signals. The model also included:

- Batch Normalization
- MaxPooling
- Dropout

Finally, a Softmax layer was used to classify sleep stages into five classes.

8. Callbacks configuration

Several callbacks were used to improve the training process:

- EarlyStopping to stop training when performance stopped improving
- ReduceLROnPlateau to reduce the learning rate automatically
- ModelCheckpoint to save the best model

9. Model training

The model was trained using the balanced dataset for 30 epochs, while validation data was used to monitor performance during training.

10. Roc curve & confusion matrix

ROC Curve and AUC were used to evaluate the model's ability to distinguish between different sleep stage classes.

A Confusion Matrix was used to analyze the model's performance by identifying correctly classified classes and the classes that were confused with each other.